Identification of CP and Tucker Factor Models

Sangmyung Ha

January 1, 2026

1 Identification of CP and Tucker decompositions

1.1 CP Decomposition

Let X be a $N \times M \times T$ dimensional cube given by

$$X = \sum_{k=1}^{K} (\alpha_k \circ \beta_k \circ f_k). \tag{1}$$

Under general assumptions, e.g., Jiang and Sidiropoulos (2004), $(\alpha_k \circ \beta_k \circ f_k)_{k=1}^K$ are identified up to permutations of indices and scale changes. A sufficient condition for identification of $(\alpha_k, \beta_k, f_k)_{k=1}^K$ is that $(\alpha_k), (\beta_k)$, and (f_k) are linearly independent. Given X, the columns $(\alpha_k, \beta_k, f_k)_{k=1}^K$ can be recovered up to the identified component by solving the minimization problem:

$$\min_{(a_k, b_k, c_k)_1^K} \left|\left|\left| X - \sum_{k=1}^{K} (a_k \circ b_k \circ c_k) \right|\right|\right|.$$

1.2 Tucker Decomposition

Let X be given by a $N \times M \times T$ dimensional cube. We define the multilinear rank of X as a tuple given by

$$\text{m-rank}(X) = (\dim(\text{span}(X_\alpha)), \dim(\text{span}(X_\beta)), \dim(\text{span}(X_f)))$$

Let X be given by:

$$X = \sum_{k_1, k_2, k_3 = 1}^{K_1, K_2, K_3} w_{k_1 k_2 k_3} (\alpha_{k_1} \circ \beta_{k_2} \circ f_{k_3}). \tag{2}$$

We denote $W = (w_{k_1 k_2 k_3})_{k_1, k_2, k_3 = 1}^{K_1, K_2, K_3} \in \mathbb{R}^{K_1 \times K_2 \times K_3}$ the 'core' tensor and $A \in \mathbb{R}^{N \times K_1}, B \in \mathbb{R}^{M \times K_2}$, and $F \in \mathbb{R}^{T \times K_3}$ the matrices with $(\alpha_k)_{k=1}^{K_1}$, $(\beta_k)_{k=1}^{K_2}$, and $(f_k)_{k=1}^{K_3}$ as their columns respectively. We may express X in terms of multilinear matrix multiplication as

$$(A, B, F) \cdot W := \sum_{k_1, k_2, k_3 = 1}^{K_1, K_2, K_3} w_{k_1 k_2 k_3} (\alpha_{k_1} \circ \beta_{k_2} \circ f_{k_3})$$

The multilinear matrix multiplication is a higher-order generalization of matrix multiplication, where for two dimensional case, we have

$$(A, F) \cdot W = AWF'$$

for $A \in \mathbb{R}^{N \times K_1}, F \in \mathbb{R}^{T \times K_2}$, and $W \in \mathbb{R}^{K_1 \times K_2}$. Note that multilinear matrix multiplication performs the actions $A : \mathbb{R}^{K_1} \to \mathbb{R}^N$, $B : \mathbb{R}^{K_2} \to \mathbb{R}^M$, and $F : \mathbb{R}^{K_3} \to \mathbb{R}^T$ along each of dimensions of W. Thus, we have

$$\begin{aligned}(A, B, F) \cdot W &= (AP^{-1}P, BQ^{-1}Q, FR^{-1}R) \cdot W \\ &= (AP^{-1}, BQ^{-1}, FR^{-1}) \cdot (P, Q, R) \cdot W \\ &= (\tilde{A}, \tilde{B}, \tilde{F}) \cdot \tilde{W},\end{aligned}$$

where $\tilde{A} = AP^{-1}, \tilde{B} = BQ^{-1}, \tilde{F} = FR^{-1}$ and $(P, Q, R) \cdot W$ for any invertible matrices $P \in \mathbb{R}^{K_1 \times K_1}, Q \in \mathbb{R}^{K_2 \times K_2}, R^{K_3 \times K_3}$. Here, we use the associativity of matrix multiplication on the second equality. We may write $(AP^{-1}, BQ^{-1}, FR^{-1}) \cdot (P, Q, R) \cdot W = (AP^{-1}, BQ^{-1}, FR^{-1}) \cdot ((P, Q, R) \cdot W)$ to be more clear which action is done first. Thus, given X, (W, A, B, F) are not separately identified. The identifying restriction Babii et al. (2025) uses are the following:

Assumption 1. *The columns of A, B, and F are orthonormal, i.e.,*

$$A'A = I_{K_1}, \quad B'B = I_{K_2}, \quad F'F = I_{K_3}$$

and the matricized core tensor W has orthogonal rows, i.e.,

$$W_\alpha W_\alpha' = D_1, \quad W_\beta W_\beta' = D_2, \quad W_f W_f' = D_3$$

where D_1, D_2, and D_3 are $K_1 \times K_1$, $K_2 \times K_2$, and $K_3 \times K_3$ dimensional diagonal matrices with decreasing strictly positive diagonal elements and W_α, W_β, and W_f are the matricized W, e.g., $(k_1, (k_2 - 1)K_3 + k_3)$-th element of W_α is given by $w_{k_1 k_2 k_3}$.

By matricizing X, we obtain the three expressions:

$$\begin{aligned}X_\alpha &= AW_\alpha(B \otimes F)' \\ X_\beta &= BW_\beta(F \otimes A)' \\ X_f &= FW_f(A \otimes B)'.\end{aligned}$$

Under Assumption 1, A, B, and F are identified as the left singular vectors of X_α, X_β, and X_f respectively, up to sign changes. To see this, note that

$$X_\alpha X_\alpha' = AW_\alpha(B \otimes F)'(B \otimes F)W_\alpha' A' = AD_1A'$$

where columns of A are orthonormal. Thus, (A, B, F) can be obtained as the left singular vectors of X_α, X_β, and X_f respectively without any iterations. After obtaining (A, B, F), we may obtain W by

$$(A', B', F') \cdot X = (A', B', F') \cdot (A, B, F) \cdot W = W$$

Remark 1:
The identification restrictions in Assumption 1 allows (W, A, B, F) to be recovered after only

performing three singular value decompositions. Note that assuming W to have multilinear rank given by (K_1, K_2, K_3) is crucial. Otherwise, diagonal matrices D_1, D_2, D_3 cannot all be strictly positive. If any diagonal elements are zero, (W, A, B, F) will not be identified. To see this, suppose the last diagonal element of D_1 is zero. Then, we have

$$X_\alpha X_\alpha' = AD_1A' = \sum_{k=1}^{K_1-1} d_{kk}\alpha_k\alpha_k'$$

so that only the first $K_1 - 1$ columns of A are identified.

Remark 2:
Consider the two dimensional case. Let $X \in \mathbb{R}^{N\times T}$ be given by

$$X = \sum_{k_1,k_2=1}^{K_1,K_2} w_{k_1k_2}(\alpha_{k_1} \circ f_{k_2}) = (A, F) \cdot W, \tag{3}$$

where $W = (w_{k_1k_2})_{k_1,k_2=1}^{K_1,K_2} \in \mathbb{R}^{K_1\times K_2}$ and $A \in \mathbb{R}^{N\times K_1}, F \in \mathbb{R}^{T\times K_2}$. As in the three dimensional case, we have

$$X = (A, F) \cdot W = (AP^{-1}, FQ^{-1}) \cdot (P, Q) \cdot W = (\tilde{A}, \tilde{F}) \cdot \tilde{W}$$

for any invertible matrices $P \in \mathbb{R}^{K_1\times K_1}, Q \in \mathbb{R}^{K_2\times K_2}$. However, we may not assume

$$A'A = I_{K_1}, \quad F'F = I_{K_2}$$
$$WW' = D_1, \quad W'W = D_2$$

where D_1 and D_2 are $K_1 \times K_1$ and $K_2 \times K_2$ dimensional diagonal matrices with decreasing strictly positive diagonal elements unless $K_1 = K_2$. This is because for the case $K_1 \neq K_2$, W is either not full column rank or full row rank. Thus, for (W, A, F) in equation (3) to be identified, we need $K_1 = K_2 = K$. In this case, we have

$$X = (A, F) \cdot W = AWF',$$

where we may identify (W, A, F) by assuming A, F are left and right singular vectors and W is diagonal elements with decreasing singular values. Suppose we assume

$$A'A = I_K, \quad F'F = I_K$$
$$WW' = D_1, \quad W'W = D_2$$

where D_1 and D_2 are $K \times K$ and $K \times K$ dimensional diagonal matrices with decreasing strictly positive diagonal elements. Then, we have $D_1 = D_2 = D$ because non-zero eigenvalues of WW' and $W'W$ are the same. This implies (A, F) are identified as left and right singular vectors and $D^{1/2}$ are the singular values of X.

Remark 3
It is also common to model $N \times M$ dimensional time series $x_t, t = 1 \ldots, T$ of slices of $X = (x_t)_{t=1}^T \in \mathbb{R}^{N\times M\times T}$ as

$$x_t = \sum_{k_1,k_2}^{K_1,K_2} w_{k_1k_2,t}(\alpha_{k_1} \circ \beta_{k_2}) \tag{4}$$

This implies that X is given by:

$$X = \sum_{k_1,k_2,s=1}^{K_1,K_2,T} w_{k_1k_2,t}(\alpha_{k_1} \circ \beta_{k_2} \circ e_t) = (A, B, I_T) \cdot W,$$

where $W = (w_{k_1k_2,t})_{k_1,k_2,t=1}^{K_1,K_2,T} \in \mathbb{R}^{K_1 \times K_2 \times T}$. Thus, modeling the slices (x_t) in Tucker form as in Equation (4) implies using elementary basis functions for the loadings in the third direction, i.e., restricting $K_3 = T$ and $F = I_T$ in the model (2).

2 Identification of the Common Component

In practice, the observations X do not have exact CP or Tucker decomposition as in equation (1) or (2). It is more common to model X as a sum of the common component and error term as in

$$X = \sum_{k=1}^{K}(\alpha_k \circ \beta_k \circ f_k) + E$$

or

$$X = \sum_{k_1,k_2,k_3=1}^{K_1,K_2,K_3} w_{k_1k_2k_3}(\alpha_{k_1} \circ \beta_{k_2} \circ f_{k_3}) + E,$$

where $E = (\varepsilon_{ijt})_{i,j,t=1}^{N,M,T}$.

There are two main approaches in the literature for identifying the common component from the idiosyncratic errors E.

First, one may rely on the pervasiveness assumption, as in Chamberlain and Rothschild (1983), Bai and Ng (2002), and Stock and Watson (2002). Under this approach, the elements of E are required to have weak serial and cross-sectional dependence so that the spectral norm of E is $O_p(\sqrt{N+M+T})$. At the same time, the factor loadings and factors must be pervasive in the sense that all the nonzero (generalized) singular values of the common component diverge at the rate $\sqrt{NMT}$, ensuring that the common component is asymptotically distinguishable from the idiosyncratic errors. One may also allow slower divergence rates for the common component provided that all nonzero singular values of the common component still diverge faster than $O_p(\sqrt{N+M+T})$; see, for example, Bai and Ng (2023).

Second, the common component may be identified by assuming that the idiosyncratic errors (ε_t) are serially uncorrelated and uncorrelated with the past common component, as in Lam and Yao (2012). Under these conditions, multiplying $x_t \in \mathbb{R}^{N \times M}$ by its lagged elements $x_{ij,t-h}$, $h > 0$, and taking expectation eliminates the idiosyncratic errors. This identifies the factor loadings (A, B) provided that the factors (f_t) exhibit nonzero serial correlation.

CP decomposition. Suppose the common component is modeled using a CP decomposition,

$$x_t = \sum_{k=1}^{K}(\alpha_k \circ \beta_k) f_{tk} + \varepsilon_t.$$

Then

$$\mathbb{E}[x_t x_{ij,t-h}] = \mathbb{E}\left[\left(\sum_{k=1}^{K}(\alpha_k \circ \beta_k) f_{tk}\right) x_{ij,t-h}\right] = ADB', \tag{5}$$

where D is diagonal with diagonal entry $D_{kk} = \mathbb{E}[f_{tk}\, x_{ij,t-h}]$. Hence A and B are identified (up to sign) as the left and right singular vectors of $\mathbb{E}[x_t x_{ij,t-h}]$ for any i and j. Given the factor loadings (A, B), we may recover f_t from x_t asymptotically by

$$(A'A)^{-1}A'x_t B(B'B)^{-1} = \mathrm{Diag}(f_t) + (A'A)^{-1}A'\varepsilon_t B(B'B)^{-1},$$

where $\mathrm{Diag}(f_t)$ is $K \times K$ dimensional diagonal matrix with elements given by f_t and the second term on the right hand side vanishes asymptotically as long as elements in (ε_t) is independent of elements in (A, B).

Tucker decomposition. Now suppose the common component follows a Tucker decomposition:

$$x_t = \sum_{k_1,k_2,k_3=1}^{K_1,K_2,K_3} w_{k_1k_2k_3}\, f_{tk_3}\, (\alpha_{k_1} \circ \beta_{k_2}) + \varepsilon_t.$$

Then

$$\mathbb{E}[x_t x_{ij,t-h}] = \mathbb{E}[(A, B, f_t') \cdot (x_{ij,t-h} W)] = A\tilde{W}B', \tag{6}$$

where the (k_1, k_2)-entry of $\tilde{W}$ is

$$\tilde{W}_{k_1k_2} = \mathbb{E}\left[\sum_{k_3=1}^{K_3} w_{k_1k_2k_3} f_{tk_3}\, x_{ij,t-h}\right].$$

Thus, A and B are identified as the left and right singular vectors of $\mathbb{E}[x_t x_{ij,t-h}]$, up to multiplication by invertible matrices. Under the identifying restrictions given in Assumption 1, (A, B) are identified up to sign changes. If we further restrict $K_3 = T$ and $F = I_T$, all of (A, B, F) are identified and we may recover W from X by

$$(A', B', I_T') \cdot X = W + (A', B', I_T') \cdot E$$

where $E = (\varepsilon_{ijt}) \in \mathbb{R}^{N\times M\times T}$ and the second term $(A', B', I_T') \cdot E = \sum_{t=1}^{T} A'\varepsilon_t B$ vanishes asymptotically, after scaling by the sample sizes NMT, as long as elements of (ε_t) are independent of elements in (A, B).

In practice, the expectations in Equations (5) and (6) may be replaced by sample averages, so only $T \to \infty$ is required for consistent estimation of (A, B). For the estimation of (f_t) in the CP model and $(w_{k_1k_2k_3})$ in the Tucker model with the restriction $F = I_T$, consistency requires that either $N \to \infty$ or $M \to \infty$, but not necessarily both. In the literature, more complex and sophisticated estimation procedures have been proposed.

3 Literature Review

Tensor factor models have been primarily studied in the statistics literature, where the common component is identified from the idiosyncratic errors by assuming that the error terms are serially uncorrelated and uncorrelated with the past common component, as in Lam and Yao (2012).

For example, Chang et al. (2023) and Han et al. (2024b) study tensor factor models based on the CP decomposition. Chang et al. (2023) propose an estimation procedure based on a generalized eigenanalysis constructed from the serial dependence structure of the underlying process. Han et al. (2024b) develop a new iterative simultaneous orthogonalization (ISO) procedure with a composite PCA initialization based on autocovariance matrices.

Similarly, Chen et al. (2022) and Han et al. (2024a) study tensor factor models based on the Tucker decomposition under the restriction $K_3 = T$ and $F = I_T$. Chen et al. (2022) introduce the TOPUP and TIPUP estimators, which are constructed using sample autocovariance matrices, while Han et al. (2024a) extend these methods using iterative orthogonal projections, yielding the iTOPUP and iTIPUP procedures.

More recently, tensor factor models whose common component is identified via the pervasiveness assumption—following the spirit of Chamberlain and Rothschild (1983) and Bai and Ng (2002)—have been developed. For instance, Chen et al. (2024) study a CP tensor factor model estimated in a manner similar to Han et al. (2024a), but based on covariance matrices rather than autocovariance matrices. Babii et al. (2025) investigate the Tucker tensor factor model without the restrictions $K_3 = T$ and $F = I_T$, employing an alternating least squares algorithm. Although the pervasiveness framework of Chamberlain and Rothschild (1983) and Bai and Ng (2002) generally allows weak serial correlation in (ε_t), both Chen et al. (2024) and Babii et al. (2025) impose zero serial correlation for technical reasons.

Our model is closest in spirit to Chen et al. (2024). The two main differences are that we use a simple alternating least squares algorithm for estimation and that we allow for serial correlation in (ε_t).

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